Time Series. Estimation, validation and forecasting

Modelització Estadística

Grau en Intel·ligència Artificial

Course 2024-25

Outline

- Estimation of parameters
- Validation of the model. Residual analysis
    - Homoscedasticity
    - Normality
    - Independence
- Forecasting with ARIMA models
- Forecasting. Examples

Estimation Criteria

Likelihood of the model

Let $X = \{x_t\}_{t=0}^{N}$ a time series with conditional **gaussian distribution**

$$L(\phi, \theta, \sigma_z^2; X) = f_{(X_1, \dots X_T)}(x_1, \dots x_T) = f_1(x_1) \sum_{i=2}^{T} f(x_i | x_{i-1}, \dots, x_1)$$

Example

For example, for an AR(1) model:

$$X_i = \phi X_{i-1} + Z_t \quad Z_t \sim N(0, \sigma_Z^2)$$

$$E(X_i | X_{i-1}, \dots, X_1) = \phi X_{i-1}$$

$$V(X_i | X_{i-1}, \dots, X_1) = \sigma_Z^2$$

$$L(\phi, \sigma_z^2; X) = f_1(x_1) \sum_{i=2}^{N} f\left(\frac{x_i - \phi x_{i-1}}{\sigma_Z}\right)$$

Conditional density: it depends on the distribution considered for the first observation (f_1)

Estimation Criteria

Maximum likelihood estimation

$$\hat{\Lambda}_{ML} = (\hat{\phi}, \hat{\theta}, \hat{\sigma}_z^2) = argmax\left(L(\phi, \theta, \sigma_z^2; X)\right)$$

Maximum likelihood estimators are **consistent** and **asymptotically eficient and gaussian**.

$$\hat{\Lambda}_{ML} \approx N(\Lambda, I_\Lambda^{-1})$$

where the Information Matrix (I_Λ) is:

$$I_\Lambda = E\left(-\frac{\partial^2 logL(\Theta; X)}{\partial \Theta^2}\right)$$

Estimation Criteria. Example

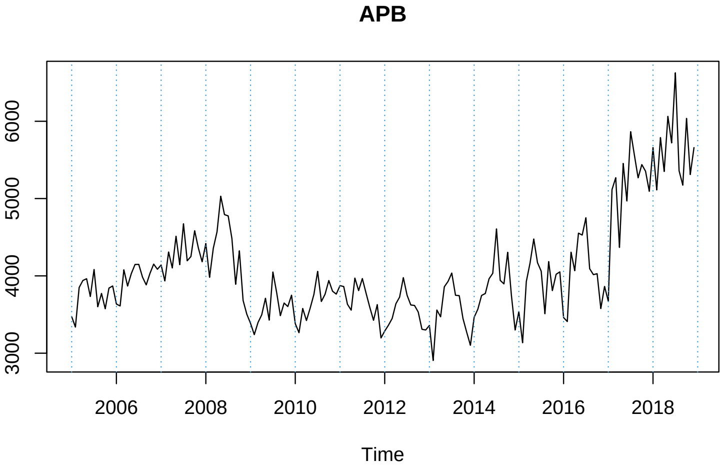
APB: Port traffic in the port of Barcelona (millions of tones)

Source: Ministry of Public Works of Spain (http://www.fomento.gob.es/BE/?nivel=2&orden=04000000)

##		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
##	2005	3473	3338	3850	3940	3963	3733	4082	3600	3773	3575	3842	3870
##	2006	3636	3611	4079	3869	4030	4148	4149	3984	3884	4032	4152	4086
##	2007	4141	3934	4309	4102	4512	4145	4674	4195	4250	4584	4359	4182
##	2008	4418	3981	4357	4570	5030	4792	4776	4482	3891	4324	3682	3504
##	2009	3385	3240	3397	3497	3710	3428	4050	3785	3486	3649	3603	3750
##	2010	3382	3265	3578	3423	3583	3760	4058	3669	3763	3940	3802	3764
##	2011	3874	3862	3633	3556	3971	3810	3965	3776	3593	3425	3628	3197
##	2012	3286	3362	3449	3639	3726	3977	3750	3623	3618	3532	3309	3300
##	2013	3359	2906	3559	3471	3859	3929	4036	3749	3744	3449	3271	3103
##	2014	3464	3569	3747	3772	3962	4033	4607	3942	3896	4305	3749	3300
##	2015	3536	3135	3927	4166	4477	4171	4061	3510	4185	3810	4021	4052
##	2016	3460	3410	4306	4067	4553	4528	4752	4092	4016	4028	3577	3863
##	2017	3674	5116	5270	4368	5454	4969	5865	5564	5268	5440	5353	5094
##	2018	5659	5112	5789	5350	6063	5720	6627	5360	5173	6037	5311	5659

Estimation Criteria. Example

APB: Port traffic in the port of Barcelona (millions of tones)



Estimation Criteria. Example. R

Proposed transformations: $W_t = (1-B)(1-B^{12})log(X_t)$

Proposed model: $ARIMA(0,1,4)(0,1,1)_{12}$

```
(model <- arima(serie,
                order = c(0,1,4),
                seasonal = list(order=c(0,1,1), period=12)))
```

```
##
## Call:
## arima(x = serie, order = c(0, 1, 4), seasonal = list(order = c(0, 1, 1), period = 12))
##
## Coefficients:
##           ma1      ma2     ma3      ma4     sma1
##       -0.4570  -0.0081  0.0405  -0.1574  -0.9227
## s.e.   0.0823   0.0946  0.0887   0.0861   0.1530
##
## sigma^2 estimated as 0.004536:  log likelihood = 187.27,  aic = -362.54
```

T-ratios: $t = \hat{\psi}/se(\hat{\psi})$

```
round(model$coef/sqrt(diag(model$var.coef)),2)
```

```
##    ma1   ma2   ma3   ma4  sma1
## -5.55 -0.09  0.46 -1.83 -6.03
```

Significant?: $|t| = |\hat{\psi}/se(\hat{\psi})| > 2$?

```
abs(model$coef/sqrt(diag(model$var.coef))) > 2
```

```
##    ma1   ma2   ma3   ma4  sma1
##   TRUE FALSE FALSE FALSE  TRUE
```

Validation of the model. Assumptions

The general expression of an ARIMA model $ARIMA(p, d, q)(P, D, Q)_s$ is:

$$\phi_p(B)\Phi_P(B^S)\left((1-B)^d(1-B^s)^D X_t - \mu\right) = \theta_q(B)\Theta_Q(B^S)Z_t$$

Random part

For all ARIMA model, the random part of the model is $Z_t \sim N(0, \sigma_Z^2)$

Residual Analysis

Assumptions:

- **Homogeneity of variance** or **Homoscedasticity** (σ_Z^2 constant)
- **Normality** ($Z_t \sim Normal$)
- **Independence** ($\rho(k) = 0 \qquad \forall k$)

Validation: Homoscedasticity

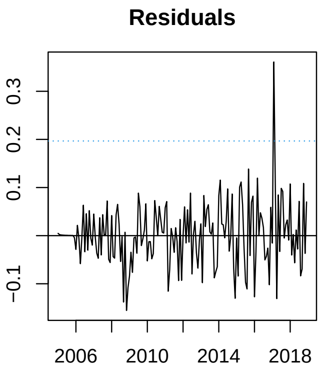
Homogeneity of variance (**Homoscedasticity**). Tools:

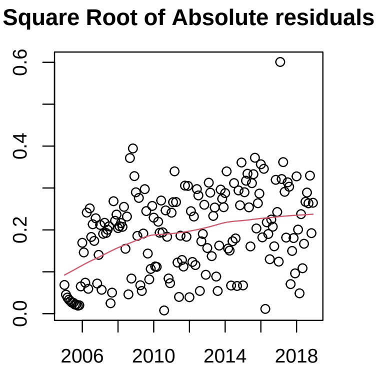
- Residuals plot
- Square root of absolute values of the residuals with smooth fit

Validation: Heterocedasticity example (KO)

```
resid <- model$residuals

par(mfrow=c(1,2), mar=c(3,3,3,3))
# Residuals plot
plot(resid,main="Residuals")
abline(h = c(0 , -3*sd(resid), 3*sd(resid)), lty = c(1,3,3), col=c(1,4,4))
# Square Root of absolute values of residuals (Homoscedasticity)
scatter.smooth(sqrt(abs(resid)), main = "Square Root of Absolute residuals", lpars = list(col=2))
```





Homogeneity of variance. Problems

- **Outlier observations**: Outlier detection and treatment
- **Volatility**: Models for the variance (GARCH)

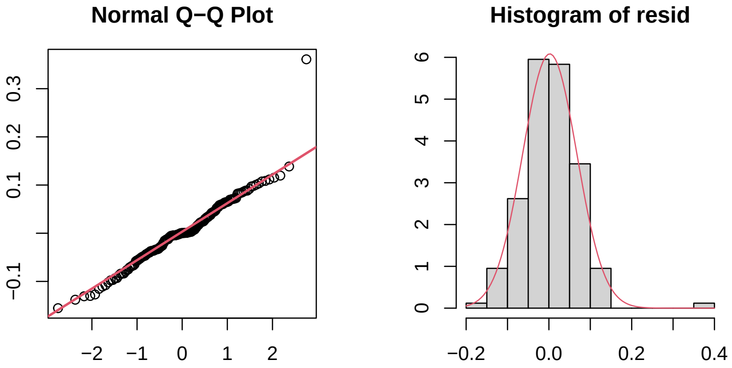
Validation: Normality

Normality. Tools:

- Quantile-Quantile plot
- Histogram with theoretical density overlapped
- Shapiro-Wilks test ($\rightarrow$ **Not recommended**. Very dependent on sample size)

Validation: Normality example (OK)

```
par(mfrow=c(1,2), mar=c(3,3,3,3))
# Normal plot of residuals
qqnorm(resid)
qqline(resid,col=2,lwd=2)
# Histogram of residuals with normal curve
hist(resid, breaks = 10, freq=F)
curve(dnorm(x, mean = mean(resid), sd = sd(resid)), col=2, add=T)
```



Normality. Problems

- **Outlier observations**: Outlier detection and treatment
- **Asymmetry or bi-modality**: Transformation, outliers or change residuals distribution
- **Heavy tails (excess of kurtosi)**: Volatility models or t-distributions for the residuals

Validation: Independence

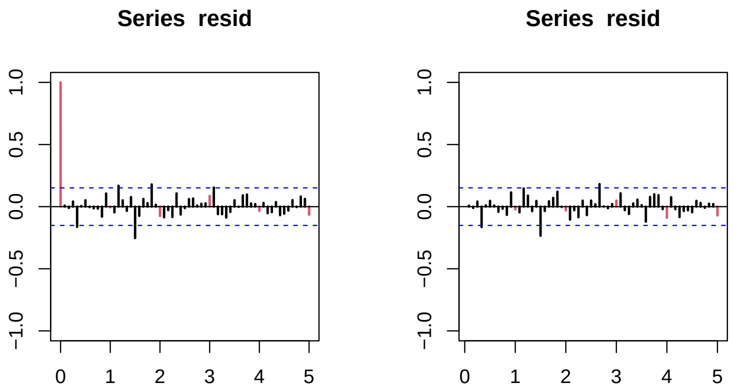
Independence. Tools:

- ACF/PACF of residuals
- LJung-Box test

Validation: Independence (OK?)

```
par(mfrow=c(1,2), mar=c(3,3,3,3))
s = 12

#ACF & PACF of residuals
par(mfrow=c(1,2))
 acf(resid, ylim=c(-1,1), lag.max=60, col=c(2, rep(1,s-1)), lwd=2)
pacf(resid, ylim=c(-1,1), lag.max=60, col=c(rep(1,s-1), 2), lwd=2)
```



Validation

Checking the significance

Checking the significance of individual autocorrelation might ignore that the configurantion of all (or a subset) of the lags may be jointly significant.

Ljung-Box Q-statistic

For a lag k test the **joint hypothesis** that the first k autocorrelations of the residuals are jointly zero:

$$H_0 : \rho_Z(1) = \cdots = \rho_Z(k) = 0$$

Test Statistic:

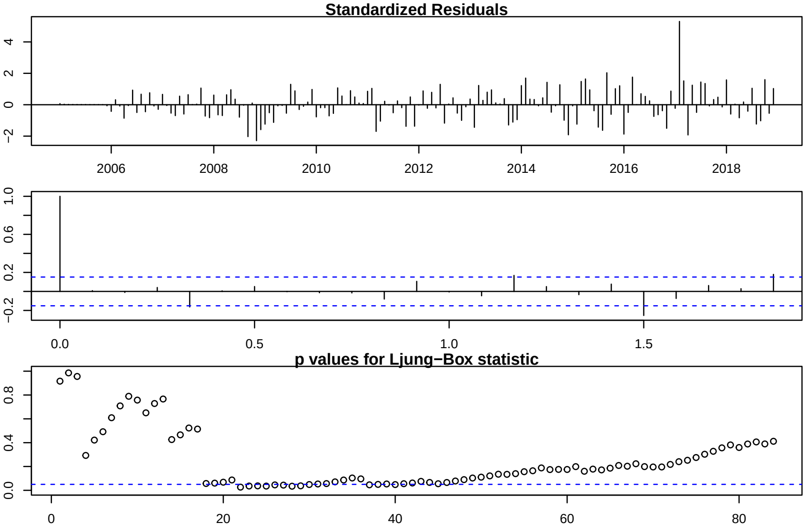
$$Q = T \sum_{i=1}^{K} \hat{\rho}_Z^2(i)$$

Asymptotic distribution:

$$Q \to \chi_k^2$$

Validation: Independence (KO)

```
par(mfrow=c(1,1),mar=c(3,3,3,3))
# Ljung-Box p-values
par(mar=c(2,2,1,1))
tsdiag(model, gof.lag = 7*12)
```



Independence. Problems

- **Significant lags**: Re-identify or add parameters to the model

Model Selection

- Information Criterion
  - Balance between godness-of-fit and simplicity of the model
  - Component for the godness-of-fit: $-2 \log L(\phi, \theta, \sigma^2 | X)$
  - Component for the simplicity: $K * p$

AIC

$$AIC = -2 \log L(\phi, \theta, \sigma^2 | X) + 2p$$

BIC

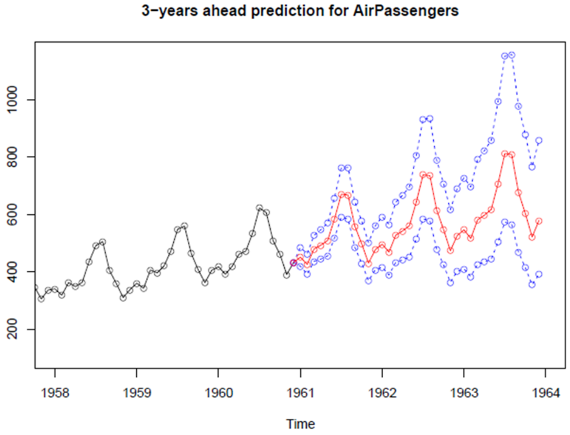
$$BIC = -2 \log L(\phi, \theta, \sigma^2 | X) + log(N)p$$

General criterion

Choose models with the **lowest AIC** or **BIC**

Forecasting and Confidence Interval (1/4)

ARIMA(0,1,1)(0,1,1)$_s$; period $s = 12$; $n.ahead = 36$



Forecasting and Confidence Interval (2/4)

Remember

A times series $\{X_t; t = 0, 1, \cdots\}$ is an **ARMA(p,q)** model if it is **stationary/causal** and **invertible**:

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_2 X_{t-p} + Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q} \qquad Z_t \sim WN(0, \sigma_Z^2)$$

with $\phi_p \neq 0$; $\theta_q \neq 0$ and $\sigma_z^2 > 0$

ARMA(p,q) model can be written in concise form as:

$$\Phi_p(B)x_t = \Theta_q(B)z_t$$

Objective

To find the best forecast for X_{t+h} at time t using the **h-step-ahead minimum mean square error predictor**, denoted by $\tilde{X}_{t+h|t}$

Forecasting and Confidence Interval (3/4)

Questions to help the construction process:

- Which is the **forecasting function**?
  i.e., Which are the **values for $\tilde{X}_{t+h|t}$** as a function of **h**?
- Which is the **memory of the model**? How are the forecast expressed as a (linear) **function of the previous** (known) observations?
- Which is the **forecasting confidence interval**?

Forecasting and Confidence Interval (4/4)

h-step-ahead-forecast

- **Previous observations** and **random noises**

  $X_{t+h} = \phi_1 X_{t+h-1} + \cdots + \phi_p X_{t+h-p} + Z_{t+h} + \theta_1 Z_{t+h-1} + \cdots + \theta_q Z_{t+h-q}$

- **AR** process of **infinite order**

  $X_{t+h} = -\pi_1 X_{t+h-1} - \pi_2 X_{t+h-2} - \cdots + Z_{t+h}$

- **MA** process of **infinite order**

  $X_{t+h} = \psi_1 Z_{t+h-1} + \psi_2 Z_{t+h-2} + \cdots + Z_{t+h}$

Forecasting example. ARMA(2,2)

Given the model

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2}$$

Suppose that **values of past observations are known**, then $Z_t = X_t - \tilde{X}_{t|t-1}$; $Z_{t-1} = X_{t-1} - \tilde{X}_{t-1|t-2}$

Substituting

$$\tilde{X}_{t+1|t} = \phi_1 X_t + \phi_2 X_{t-1} + E(Z_{t+1}) + \theta_1 Z_t + \theta_2 Z_{t-1}$$
$$\tilde{X}_{t+2|t} = \phi_1 \tilde{X}_{t+1|t} + \phi_2 X_t + E(Z_{t+2}) + \theta_1 E(Z_{t+1}) + \theta_2 Z_t$$
$$\tilde{X}_{t+3|t} = \phi_1 \tilde{X}_{t+2|t} + \phi_2 \tilde{X}_{t+1|t} + E(Z_{t+3}) + \theta_1 E(Z_{t+2}) + \theta_2 E(Z_{t+1})$$
$$\ldots$$
$$\tilde{X}_{t+h-3|t} = \phi_1 \tilde{X}_{t+h-1|t} + \phi_2 \tilde{X}_{t+h-2|t} \ , \ h > 2$$

Non Stationary ARIMA process Forecast. ARIMA(0,1,1) (1/3)

Model:

$$X_t = X_{t-1} + Z_t + \theta Z_{t-1}$$

also defined by

$$(1 - B)X_t = (1 - \theta B)Z_t$$

X_t **can be expressed as**

$$\pi(B) = \frac{1 - B}{1 + \theta B} = 1 - (\theta + 1)B + \theta(\theta + 1)B^2 - \theta^2(\theta + 1)B^3 + \cdots$$

Non Stationary ARIMA process Forecast. ARIMA(0,1,1) (2/3)

Model:

$$\tilde{X}_{t+1|t} = (\theta+1)X_t - \theta(\theta+1)X_{t-1} + \theta^2(\theta+1)X_{t-2} - \cdots$$

moving θ

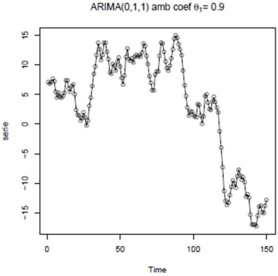
$$\tilde{X}_{t+1|t} = (\theta+1)X_t - \theta\tilde{X}_{t|t-1}$$

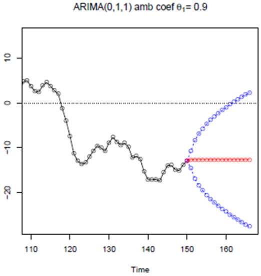
Using the parameter $\lambda = 1+\theta$. Clearly $\theta = \lambda - 1 = -(1-\lambda)$

$$\tilde{X}_{t+1|t} = \lambda X_t - (1-\lambda)\lambda X_{t-1} + (1-\lambda)^2\lambda X_{t-2} - \cdots = \lambda X_t + (1-\lambda)\tilde{X}_{t|t-1}$$

Non Stationary ARIMA process Forecast. ARIMA(0,1,1) (3/3)

Hence, the **forecast is the a linear combination**of the last observation and the forecast obtained in the previous step.





Variance of the forecasting error

Variance of the 1-step-ahead forecasting error

$$Var(e_t(1)) = E\left((X_{t+1} - \tilde{X}_{t+1|t})^2\right) = E(Z_{t+1}^2) = \sigma_Z^2$$

H step forecast for h 2 is

$$Var(e_t(h)) = \sigma_Z^2(1 + \psi_1^2 + \psi_2^2 + \cdots \psi_{h-1}^2)$$

Seasonal Processes Forecasting. ARIMA(0,1,1)(0,1,1)$_{12}$ (1/3)

Model:

$$(1 - B)(1 - B^{12})X_t = (1 + \theta B)(1 + \theta_{12} B^{12})Z_t$$

also defined by

$$(X_t - X_{t-12}) - (X_{t-1} - X_{t-13}) = Z_t + \theta_1 Z_{t-1} + \theta_{12} Z_{t-12} + \theta_1 \theta_{12} Z_{t-13}$$

Hence,

$$X_{t+1} = X_t + (X_{t-11} - X_{t-12} + Z_{t+1} + \theta_1 Z_{t-11} + \theta_1 \theta_{12} Z_{t-12})$$

Seasonal Processes Forecasting. ARIMA(0,1,1)(0,1,1)$_{12}$ (2/3)

The forecasting function:

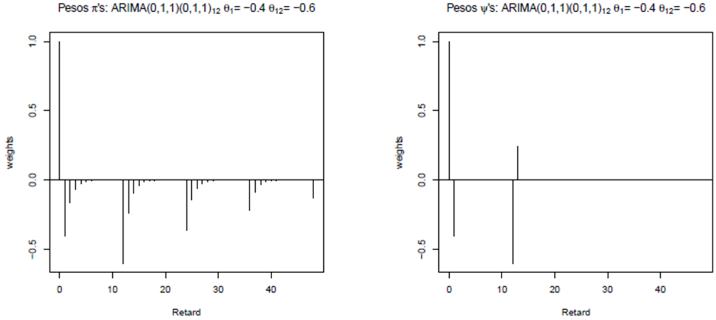
$$\tilde{X}_{t+1|t} = X_t + (X_{t-11} - X_{t-12}) + \theta_1 Z_t + \theta_1 2 Z_{t-11} + \theta_1\theta_{12} Z_{t-12}$$

$$\tilde{X}_{t+2|t} = \tilde{X}_{t+1|t} + (X_{t-10} - X_{t-11}) + \theta_{12} Z_{t-10} + \theta_1\theta_{12} Z_{t-11}$$

$$\ldots$$

$$\tilde{X}_{t+h|t} = \tilde{X}_{t+h-1|t} + (\tilde{X}_{t+h-12|t} - \tilde{X}_{t+h-13|t})\ ,\ h > 13$$

Seasonal Processes Forecasting. ARIMA(0,1,1)(0,1,1)$_{12}$ (3/3)



Weights π and ψ for the model ARIMA(0,1,1)(0,1,1)$_{12}$ for the parameters above

Schema. Recap